MINOR PROJECT REPORT

Exploratory Analysis of Financial Transactions

Submitted as part of the Data Science Internship

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1. Abstract

This report is about my minor project, *Exploratory Analysis of Financial Transactions*, which I did during my Data Science internship at Skillairo. I explored a credit card transactions dataset of 284,807 records using Python, Pandas and Matplotlib. I checked the data quality, the class imbalance, the pattern of transaction amounts and times, the correlation of each feature with fraud, and the outliers. I found that fraud is very rare (only 0.172% of transactions) and that the transaction amount alone cannot detect it, while some of the anonymized features, especially V17, V14 and V12, are strongly related to fraud.

2. Introduction

2.1 Problem Statement

Financial datasets are large and complex, and it takes careful analysis to find patterns in them. Fraudulent transactions are hidden among a very large number of normal ones.

2.2 Objectives

- To understand the transaction-level features of the dataset.
- To perform exploratory data analysis (EDA) and find trends and anomalies.
- To visualize the distribution of transactions.
- To highlight suspicious patterns related to fraud.

3. Dataset Description

I used the Credit Card Fraud Detection dataset from Kaggle (mlg-ulb/creditcardfraud). It contains transactions made by European cardholders over two days. The main details are given in Table 1.

Property	**Details**
Number of rows	284,807
Number of columns	31 (32 after I added the Hour column)
Missing values	None
Time	Seconds since the first transaction
V1 to V28	Anonymized features (PCA transformed)
Amount	Transaction amount
Class	Target: 0 = Normal, 1 = Fraud

Table 1: Details of the dataset

4. Tools and Technologies Used

- **Python 3** as the programming language.
- **Pandas** to load, inspect and summarize the data.
- **Matplotlib** to make the graphs.
- **JupyterLab** as the working environment.

5. Methodology

I followed these steps in my notebook:

- Loaded the dataset and checked its shape, data types and missing values.
- Counted the normal and fraud transactions and plotted the class distribution.
- Studied the Amount column using summary statistics, a histogram and a box plot.
- Created an Hour column from the Time column and compared normal and fraud transactions by hour.
- Found the correlation of every feature with the Class column.
- Detected outliers in Amount using the IQR method.

6. Analysis and Results

6.1 Data Inspection

First I loaded the dataset and looked at the first few rows. The dataset has 284,807 rows and 31 columns. All columns are numeric (float64) and there are no missing values, so I did not need to clean or fill any data.

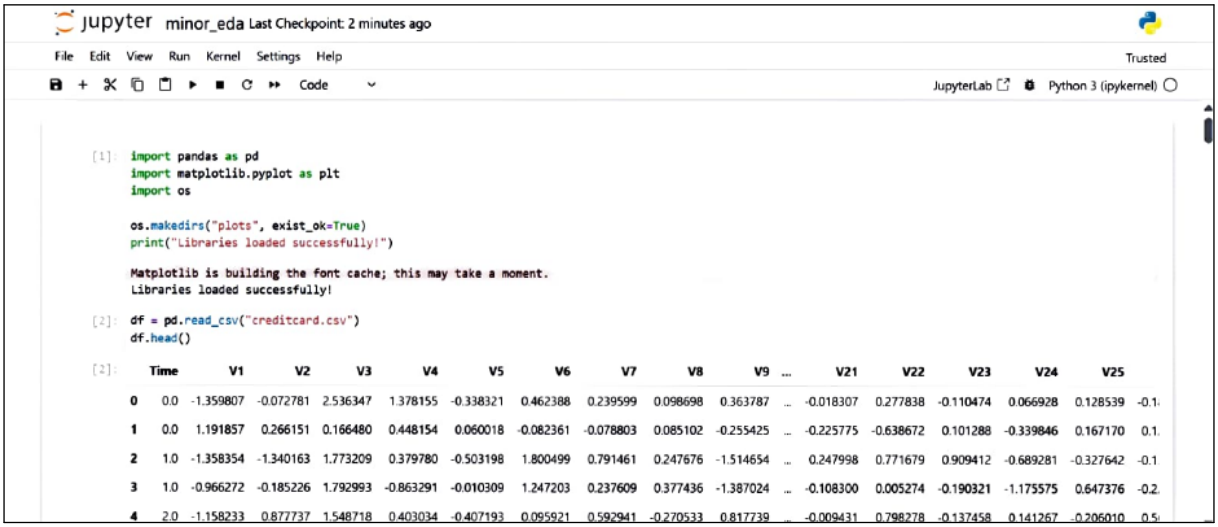


Fig. 1: Loading the dataset and viewing the first rows



Fig. 2: Shape, data types and non-null counts

6.2 Class Imbalance

Out of 284,807 transactions, 284,315 are normal (99.827%) and only 492 are fraud (0.173%). So the data is extremely imbalanced. This is important because a model that predicts every transaction as normal would still be about 99.8% accurate without finding any fraud. I used a log scale in the graph so that the small fraud bar can also be seen.



Fig. 3: Count and percentage of each class

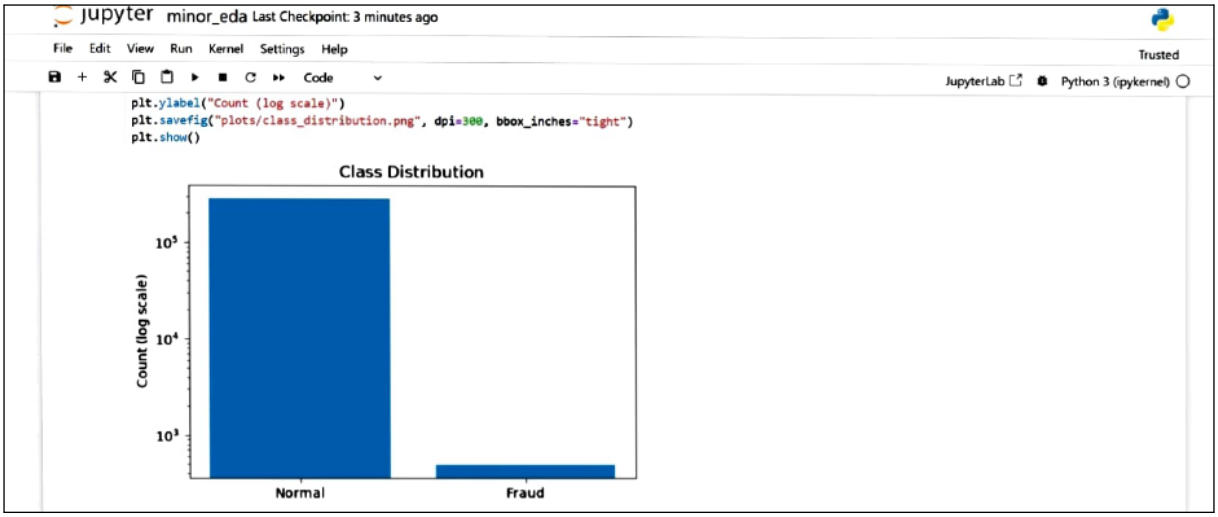


Fig. 4: Class distribution (log scale)

6.3 Transaction Amount

The average transaction amount is about 88.35, but the median is only 22.00 and the maximum is 25,691.16. Since the mean is much higher than the median, the distribution is right-skewed: most transactions are small and only a few are very large.

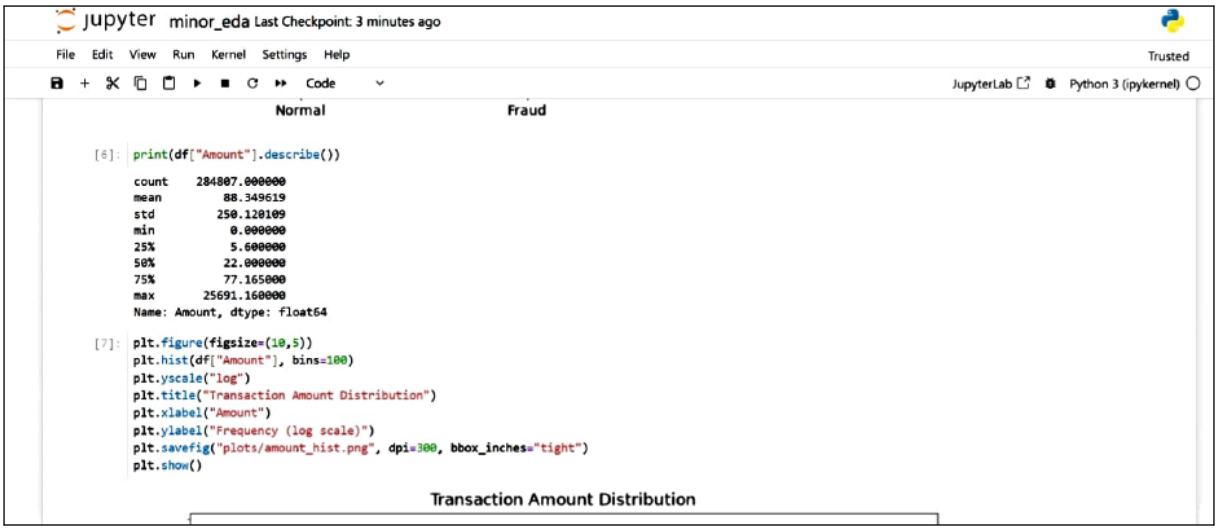


Fig. 5: Summary statistics of Amount

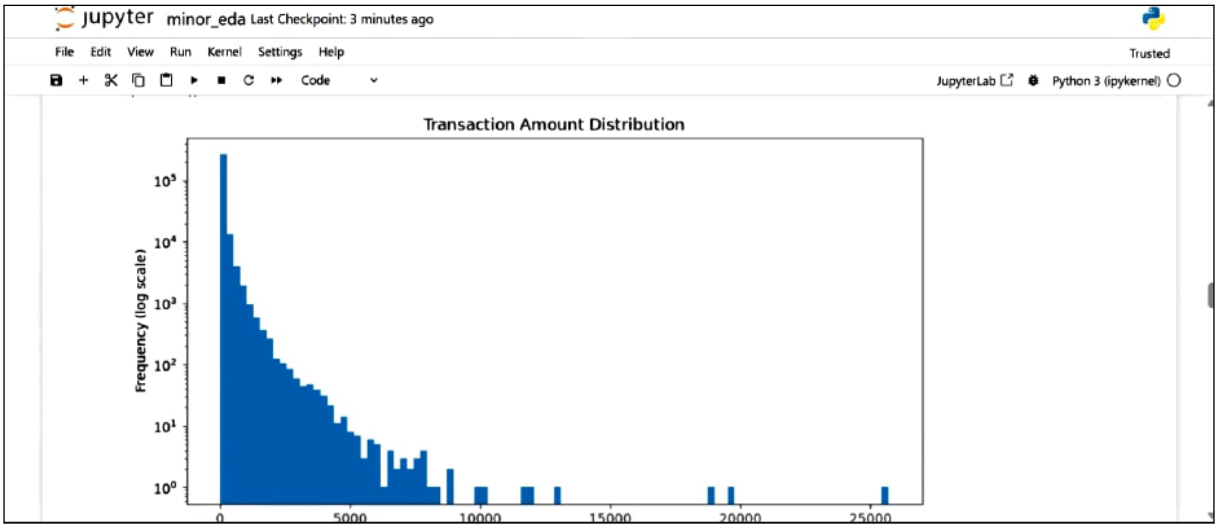


Fig. 6: Histogram of transaction amounts (log scale)

I then compared the amount of normal and fraud transactions with a box plot. The fraud amounts lie in almost the same range as the normal ones, and the largest fraud amount (about 2,000) is much smaller than the largest normal amount (above 25,000). Because the two overlap so much, the amount alone cannot be used to detect fraud.

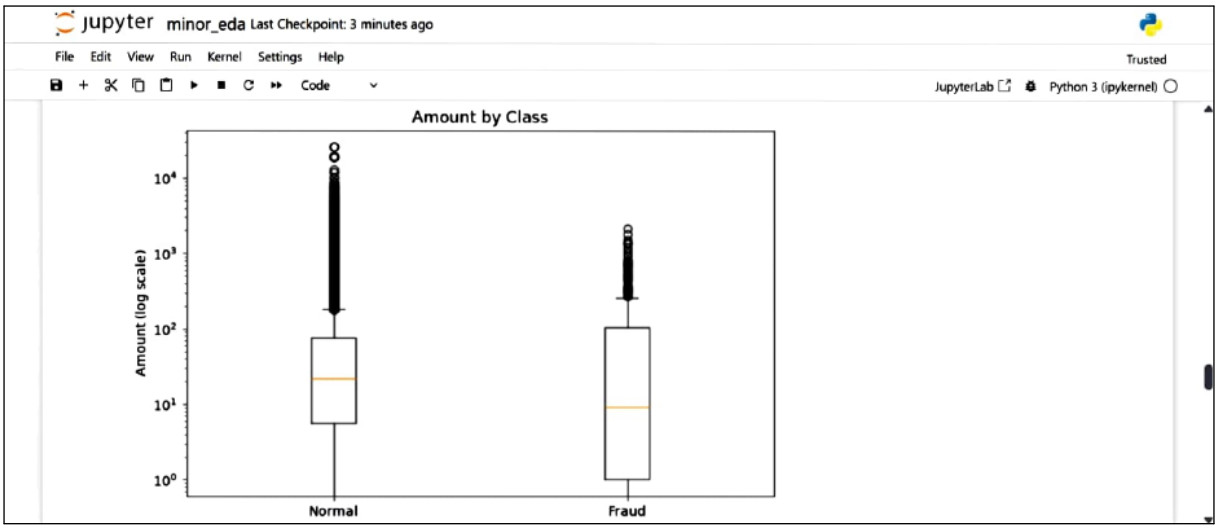


Fig. 7: Box plot of Amount by class (log scale)

6.4 Time Pattern

I made a new column called Hour using (Time / 3600) % 24 and plotted the hourly distribution for both classes. Normal transactions follow a daily cycle, with less activity in the early hours and steady activity during the day. Fraud transactions show clear peaks in the early morning (around 2 to 3 AM) and near 11 AM.

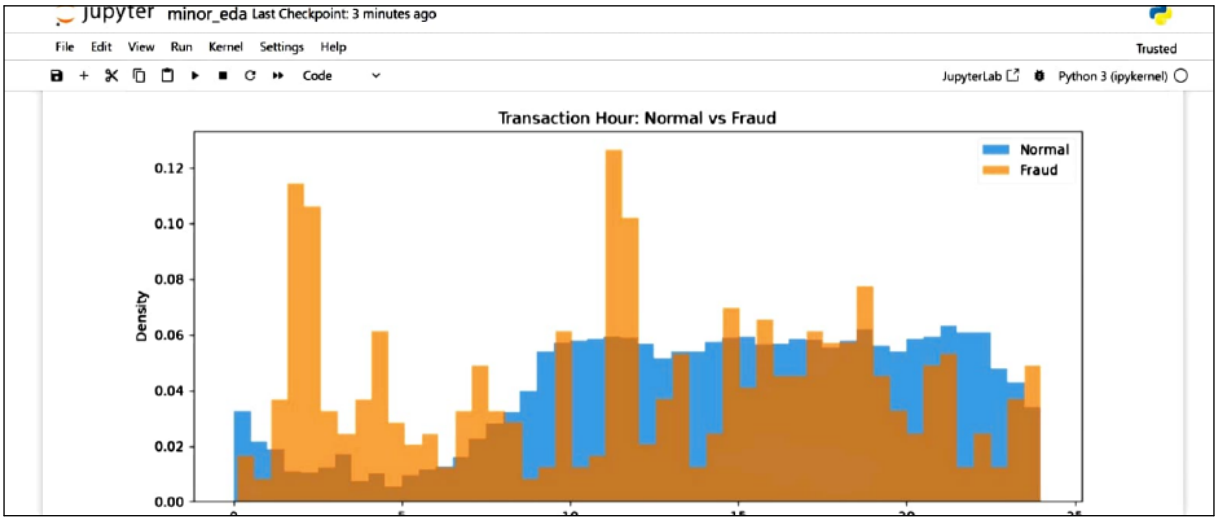


Fig. 8: Transaction hour for normal and fraud transactions

6.5 Correlation with Fraud

I sorted all the features by their correlation with Class. The strongest negative correlations are V17 (-0.326), V14 (-0.303), V12 (-0.261), V10 (-0.217) and V16 (-0.197). The strongest positive ones are V11 (0.155) and V4 (0.133). Amount (0.006) and Time (-0.012) are almost not correlated with fraud, which agrees with the box plot result.



Fig. 9: Correlation of features with Class (top part)

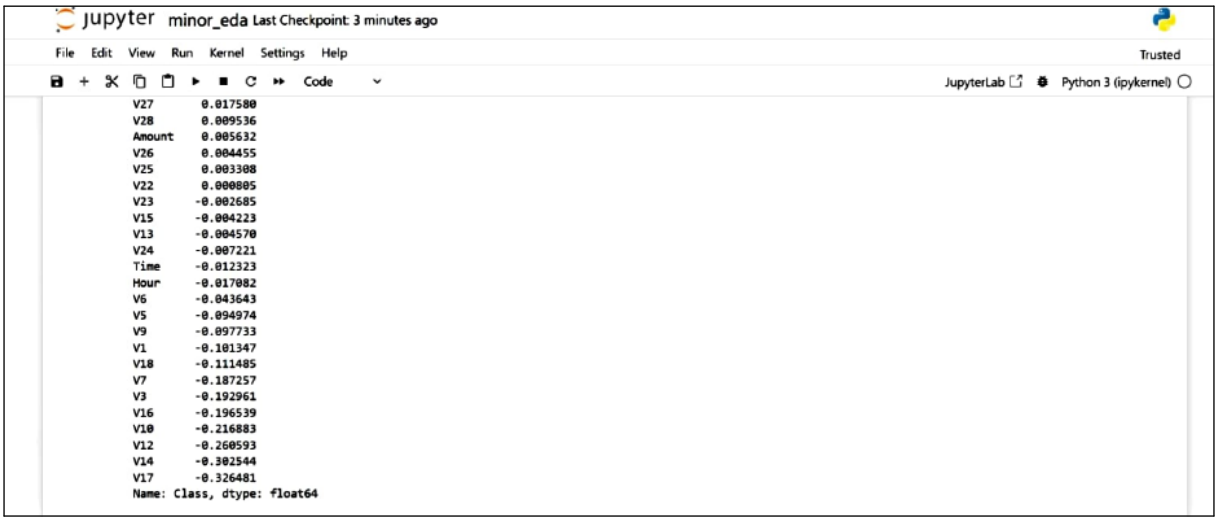


Fig. 10: Correlation of features with Class (bottom part)

6.6 Outlier Analysis

I used the IQR rule (1.5 x IQR) on the Amount column. It marked 31,904 transactions (about 11.2%) as outliers, but only 91 of them are fraud. These 91 are only about 18.5% of the 492 fraud cases, so most fraud transactions are not outliers in amount. This again shows that a large amount is not a reliable sign of fraud.



Fig. 11: Outlier detection on Amount using IQR

7. Key Findings

- **Class imbalance:** Fraud is extremely rare, only 0.172% of all transactions.
- **Amount distribution:** It is highly right-skewed, with mostly small transactions and a few very large ones.
- **Amount and class:** Fraud and normal amounts overlap a lot, so amount alone is not enough to detect fraud.
- **Time pattern:** Fraud peaks at odd hours such as 2 to 3 AM, while normal transactions follow a normal day and night cycle.
- **Correlation:** The features V17, V14 and V12 are strongly related to fraud and will be useful for detection.
- **Outliers:** There are 31,904 outliers in amount but only 91 are fraud, which confirms that amount alone is not enough.

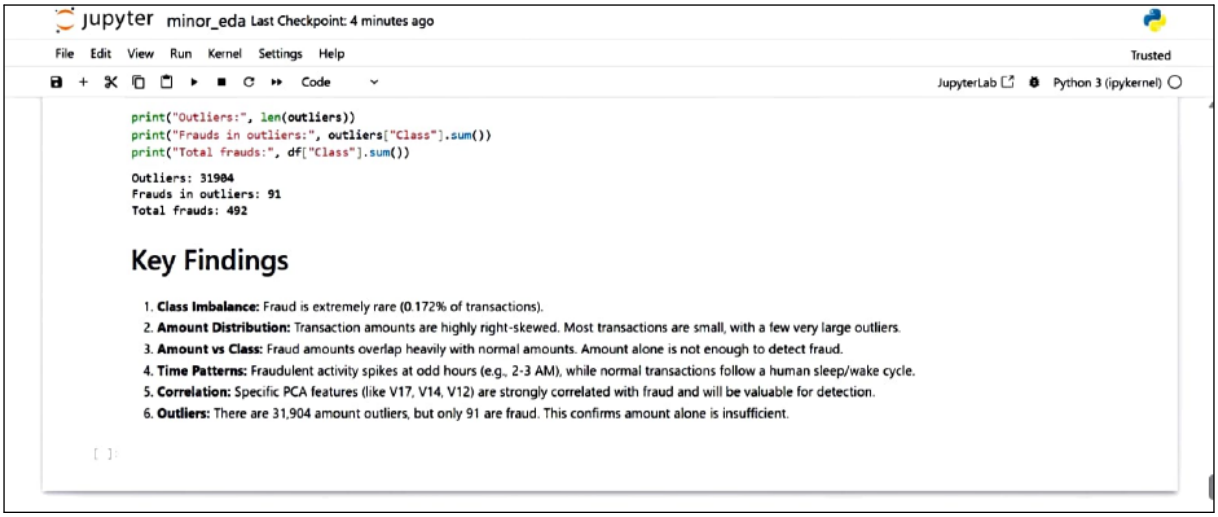


Fig. 12: Key findings written in my notebook

8. Suspicious Patterns Observed

- Transactions in the early morning hours have a higher share of fraud than normal transactions.
- Unusual values of V17, V14, V12, V10 and V16 are linked with fraud.
- Fraud mostly involves small or medium amounts, which helps it look like normal spending.

9. Limitations

- The features V1 to V28 are anonymized, so I could not explain what they mean in real life.
- There are only 492 fraud cases, so patterns like the hourly peaks may not be fully reliable.
- The Time column covers only two days and starts from the first transaction, so the Hour column is only an approximation of the actual time of day.
- Correlation only shows a relationship between features and does not prove the cause.

10. Conclusion

In this project I did an exploratory analysis of credit card transactions. I learned that fraud is very rare in the data, it cannot be found from the amount alone, it shows some unusual behaviour at certain hours, and a few features (V17, V14 and V12) separate it best. These findings will help in the next step, which is building models for fraud detection. For that I will need to handle the class imbalance and use metrics like precision and recall instead of accuracy.

11. Links

GitHub link: https://github.com/danishveersigh/Credit-Card-Fraud-Analysis

Dataset: https://www.kaggle.com/datasets/mlg-ulb/creditcardfraud